

# Predictive Analytics Group Project Assignment

**Weighting:** 50% of final module grade

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## Project Overview and Timeline

The project is divided into **two phases**: an exploratory methodology phase and a final implementation and presentation phase.

### Key Dates

- **13th March** – Datasets released
  - **24th March** – Phase 1 meeting with lecturer (50% of final mark)
  - **7th April** – Model & executive summary submission (50% of final mark)
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## Phase 1: Data Exploration and Methodology (13th March – 24th March)

During this phase, each group must:

- Explore both datasets
- Clean and prepare the data
- Reflect critically on preprocessing decisions
- Experiment with modelling strategies

On **24 March**, each group will attend a **15-minute meeting** with the lecturer (during the lecture)

### Phase 1 Meeting Requirements

- No slide presentation is required.
- You must be prepared to:
  - Guide the lecturer through your exploration notebooks
  - Show visualisations and exploratory analysis
  - Justify your data preprocessing and methodological decisions
  - Answer questions about your approach

After the meeting, groups will receive:

- Feedback on methodology (40% mark of their final group project mark)

- A project management mark (10% mark of their final group project mark) reflecting progress to date
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## Phase 2: Implementation and Final Presentation (After feedback)

Following feedback from Phase 1, groups will:

- Refine their methodology based on feedback from lecturer
- Implement their modelling strategy
- Evaluate and compare models
- Extract meaningful insights
- Prepare their final deliverables

### Final Assessment (7th April)

Each group must:

- Participate in **5–10 minutes of Q&A**
- Submit an **Executive Summary**

### Executive Summary Requirements

- Maximum **1000 words**
  - Between **5–10 pages**
  - Must include visualisations (charts, tables, or figures)
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## Marking Scheme (total 100 marks)

### 1. Project Management (25 marks)

Marks will be awarded for:

Submission of a Trello board link documenting the group's project management process, including task assignment and progress tracking.

It is the group's responsibility to ensure that the Trello board accurately reflects the project's progress, including task allocation, updates, and completion. While the project is completed in groups, individual marks for this component may vary within a team and will be determined based on evidence of each member's contribution recorded through Trello activity.

For further guidance on how Trello should be used, please refer to the lecture on 10th March.

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### 2. Methodology (40 marks)

Marks will be awarded for:

- Appropriate and well-justified data cleaning and wrangling
  - Clear preparation of the dataset for modelling
  - Exploration of multiple approaches to key tasks
  - Critical evaluation of alternative strategies
  - Ability to justify preprocessing decisions during the Phase 1 meeting
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### 3. Implementation (15 marks)

Your repository should demonstrate:

- Clean, readable, and well-documented code
  - Appropriate software engineering practices, including basic testing or validation procedures
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### 4. Executive Summary and Q&A (20 marks)

Marks will be awarded for:

- Clear and compelling insights extracted from the data
- Appropriate use and comparison of multiple models

- Transparent reporting of model performance
  - Informative and engaging visualisations
  - Professional presentation style
  - Ability to respond effectively to questions
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## General Expectations

This project assesses not only your technical skills, but also:

- Critical thinking
- Collaboration
- Communication
- Professional data science practice

You are expected to approach this project as a small data science consultancy team delivering insights to a client.

## AI Assistance Policy

Students are permitted to use AI tools to support their work.

However:

- Students are fully responsible for verifying the correctness of any AI-generated code or analysis.
- Any errors, incorrect assumptions, or methodological flaws remain the student's responsibility.
- Students must ensure they understand and can explain all submitted work.
- Blind reliance on AI without validation will result in loss of marks.

## Competition: Student Dropout Prediction Challenge (Optional)

In addition to the assessed components of this project, groups will have the opportunity to participate in a prediction competition based on the student dropout dataset.

Each group will be asked to submit a separate script specifically designed to generate dropout predictions for an unseen test dataset containing 500 students. This code will be executed during class to produce predictions.

To ensure compatibility, groups will be provided with a small sample of the test dataset in advance, allowing them to verify that their code runs correctly and produces predictions in the required format.

All submitted models will be evaluated on the hidden test dataset and ranked on a leaderboard based on prediction accuracy.

### Prizes

The London Centre for AI will sponsor two Excellence in Predictive Modeling Awards:

- 1st Prize – Highest prediction accuracy (£50 per team member)
- 2nd Prize – Second-highest prediction accuracy (£25 per team member)

These prizes are independent of the academic marking scheme and do not affect students' module grades. The cash prize is paid in John Lewis gift cards. Each prize comes with an optional LinkedIn-native digital award. The competition is intended to provide an additional opportunity for experimentation and recognition of strong predictive performance.

## Dataset: Predicting Student Dropout & Engagement in UK Online Higher Education

### Context

This dataset aggregates and anonymises the digital footprint of students participating in summer bootcamps delivered by LON.AI, a UK-based EdTech company, in partnership with several UK universities since 2018.

Each row represents a single student's journey through the online programme.

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### Stakeholder Objectives

The primary objective is:

Predicting student dropout, a significant and costly issue in online education.

A secondary objective is:

- Identifying broader patterns in student engagement
- Discovering potential student personas or behavioural clusters within the cohort

Students are encouraged to think beyond prediction and consider exploratory and unsupervised approaches where appropriate.

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## Programme Structure

The online bootcamp follows the sequence below:

1. Session 1
2. Session 2
3. Test 1
4. Session 3
5. Session 4
6. Test 2
7. Session 5
8. Test 3
9. Session 6
10. Individual coursework deadline
11. Group coursework deadline
12. Final exam (recorded as *Final grade*)

Although students may drop out at any stage, dropout often correlates with staggered tuition fee payment deadlines.

Note: Some students may receive a final grade yet still be labelled as “dropout” if they fail to pay the final tuition installment. In such cases, they are officially recorded as not having completed the programme.

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## Key Variables

### Student Characteristics

- **External:**
  - 'Y' = student is *only* enrolled in the online course
  - Otherwise, the student is also enrolled in another university programme
- **Year:** Undergraduate study year

### Engagement Metrics

- **Session 1–6:** Percentage (%) of each Zoom session viewed
- **Forum Q:** Number of question posts
- **Forum A:** Number of answer posts
- **Office hour visit:** Number of office hours attended

### Assessment Performance

- **Test 1–3:** Test grades
- **ind cw:** Individual coursework grade
- **group cw:** Group coursework grade
- **Final grade:** Final exam grade

### Target Variable

- **Dropout:**
  - 'Y' = student dropped out and did not return
  - 'N' = student completed the programme